Autoregressive Models: The Yule-Walker Equations

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Key Concepts

1) The Yule-Walker equations relate the autocorrelation sequence for a random time series to the autoregressive (AR) model parameters.

2) The relationship between the N-th order AR model parameters and the autocorrelation sequence is described by a system of N linear equations $\mathbf{Ra} = \mathbf{r}$.

    a) $\mathbf{R}$ (NxN) and $\mathbf{r}$ $(Nx1)$ are formed using the first N values of the autocorrelation sequence.

    b) $\mathbf{a}$ $(Nx1)$ is a vector containing the recursive coefficients in the N-th order AR model.

3) The white noise input power in the AR model is also a function of the autocorrelation sequence of the time series.

4) The Yule-Walker equations can be used to estimate an autoregressive model for a given time series.

    a) First estimate the autocorrelation sequence of the time series.

    b) Next solve for the AR model parameters as a function of the estimated autocorrelation sequence.

5) The Yule-Walker equations do not indicate how to choose the model order N.

Video Screenshots

- Yule-Walker equations relate AR model parameters to the autocovariance of the random process

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- Can be used to estimate AR model parameters

  - Given $x[n]$, estimate $r_{xx}[k]$
  - Solve Yule-Walker to find $a_\ell, \ell = 0, 1, \ldots N$, σ^2 from $\hat{r}_{xx}[k]$

- Does not address selection of model order N

  Usually done by balancing error against the number of parameters in the model

  - Akaike information criterion (AIC)
  - Bayesian information criterion (BIC)
  - Cross validation (CV)

Auto regressive model

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$$\sum_{k=0}^{N} a_k x[n-k] = w[n] \quad ; \quad a_0 = 1$$

multiply by $x[n-l]$ and take expectation

$$\sum_{k=0}^{N} a_k \underbrace{E\{x[n-k]x[n-l]\}}_{r_{xx}[l-k]} = \underbrace{E\{w[n]x[n-l]\}}_{\begin{cases} 0, & l>0 \\ \sigma^2, & l=0 \end{cases}}$$

Note: $$x[n-l] = -\sum_{k=1}^{N} a_k x[n-l-k] + w[n-l]$$

and $$r_{ww}[k] = \begin{cases} \sigma^2 & k=0 \\ 0 & k \neq 0 \end{cases}$$

$$\sum_{k=0}^{N} a_k r_{xx}[l-k] = \begin{cases} 0 & , \ l > 0 \\ \sigma^2 & , \ l = 0 \end{cases} \quad ; \ a_0 = 1$$

For $l > 0$:

$$\sum_{k=1}^{N} a_k r_{xx}[l-k] = -r_{xx}[l]$$

$$\begin{matrix} l=1 \\ l=2 \\ \vdots \\ l=N \end{matrix} \begin{bmatrix} r_{xx}[0] & r_{xx}[-1] & \cdots & r_{xx}[1-N] \\ r_{xx}[1] & r_{xx}[0] & \cdots & r_{xx}[2-N] \\ & & \vdots & \\ r_{xx}[N-1] & r_{xx}[N-2] & \cdots & r_{xx}[0] \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_N \end{bmatrix} = - \begin{bmatrix} r_{xx}[1] \\ r_{xx}[2] \\ \vdots \\ r_{xx}[N] \end{bmatrix}$$

Yule-Walker Equations

N equations, N unknowns

Yule-Walker equations in compact notation

$$\underline{R}\,\underline{a} = -\underline{r}$$

$$\Rightarrow \quad \underline{a} = -\underline{R}^{-1}\underline{r}$$

Given $\underline{a}$, solve for σ^2

$$\sigma^2 = \sum_{k=0}^{N} a_k r_{xx}[-k]$$

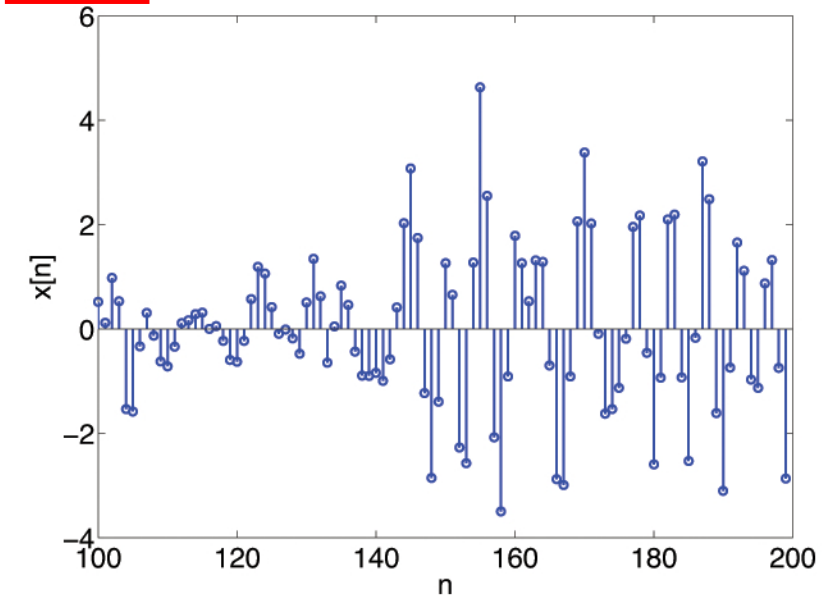
($l=0$ case of equation top of p. 3)

MATLAB: "ARYULE"

10:17

Example -

Apply aryule to $x[n]$, $n = 0, 1, \ldots, 199$

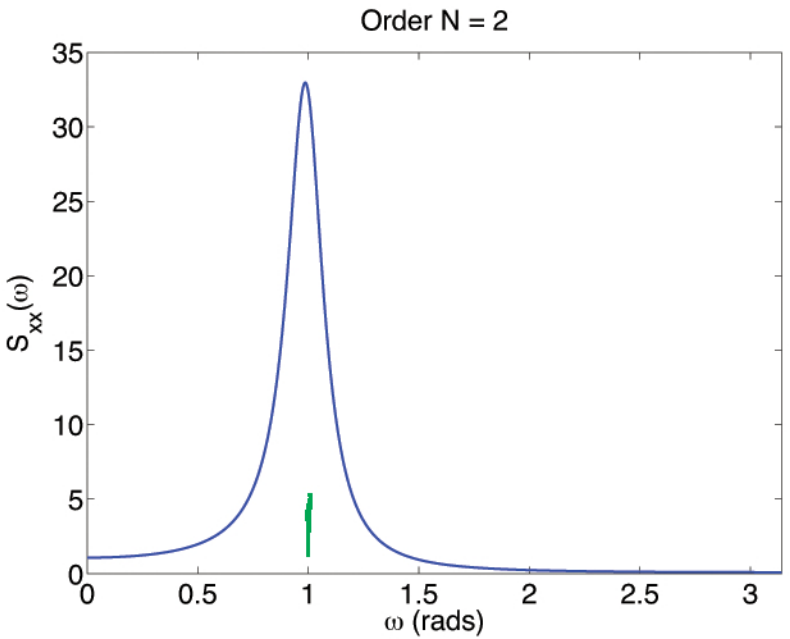


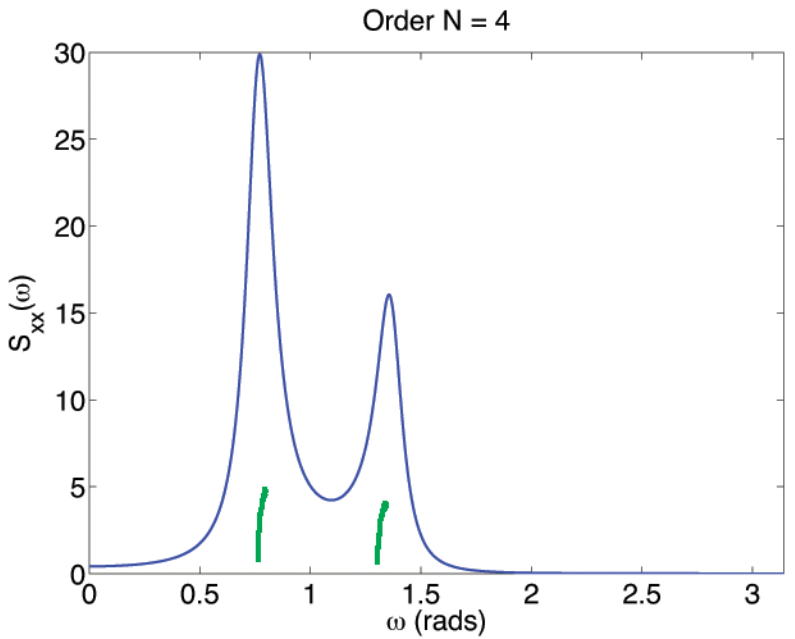
a_2 = 1 -0.9956 0.8217, sig_2 = 0.7319

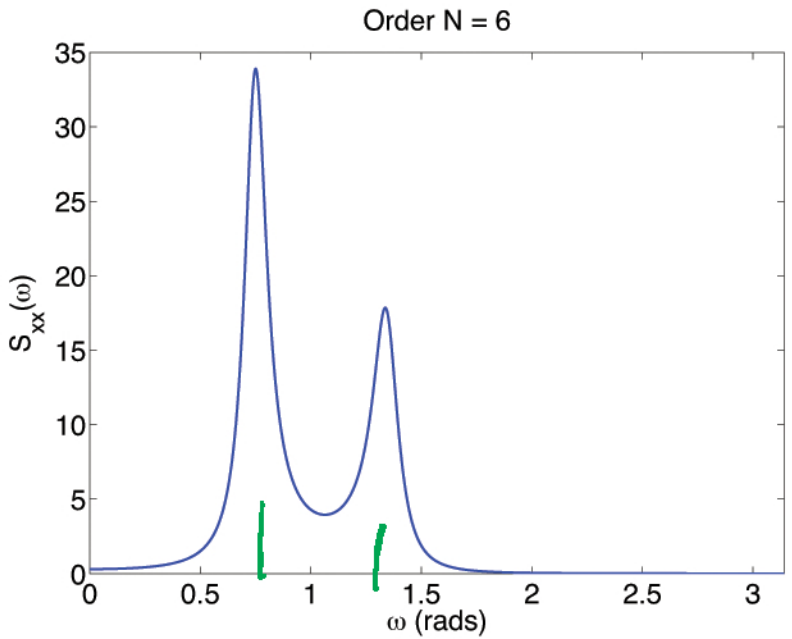
a_4 = 1 -1.7148 2.2326 -1.4833 0.7467, sig_4 = 0.2558

a_6 = 1 -1.6382 2.1337 -1.3615 0.7300 -0.0194 0.0689, sig_6 = 0.2523

$$S_{xx}(\omega) = \frac{\sigma^2}{\left| \sum_{k=0}^{N} a_k e^{-jk\omega} \right|^2}$$







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